

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459429.8784 +/- 0.0073 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.045
Transit depth (Rp/Rs)^2	2.82 +/- 1.5 [%]
Orbital Inclination (inc)	83.68 +/- 1.95
Airmass coefficient 1 (a1)	0.984 +/- 0.023
Airmass coefficient 2 (a2)	0.0098 +/- 0.008
Scatter in the residuals of the lightcurve fit is	2.33 %
Best Comparison Star	#2 - [557, 258]
Optimal Aperture	3.48
Optimal Annulus	7.21
Transit Duration (day)	0.054 +/- 0.036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.168 ± 0.045 vs 0.1394 (deviation 0.64σ)
Duration fit vs published	0.054 d vs 0.0983 d (deviation 1.23σ)
Mid-time O–C	-2.6 min
Depth SNR	3.7
Residual scatter	2.33 %

Light curve

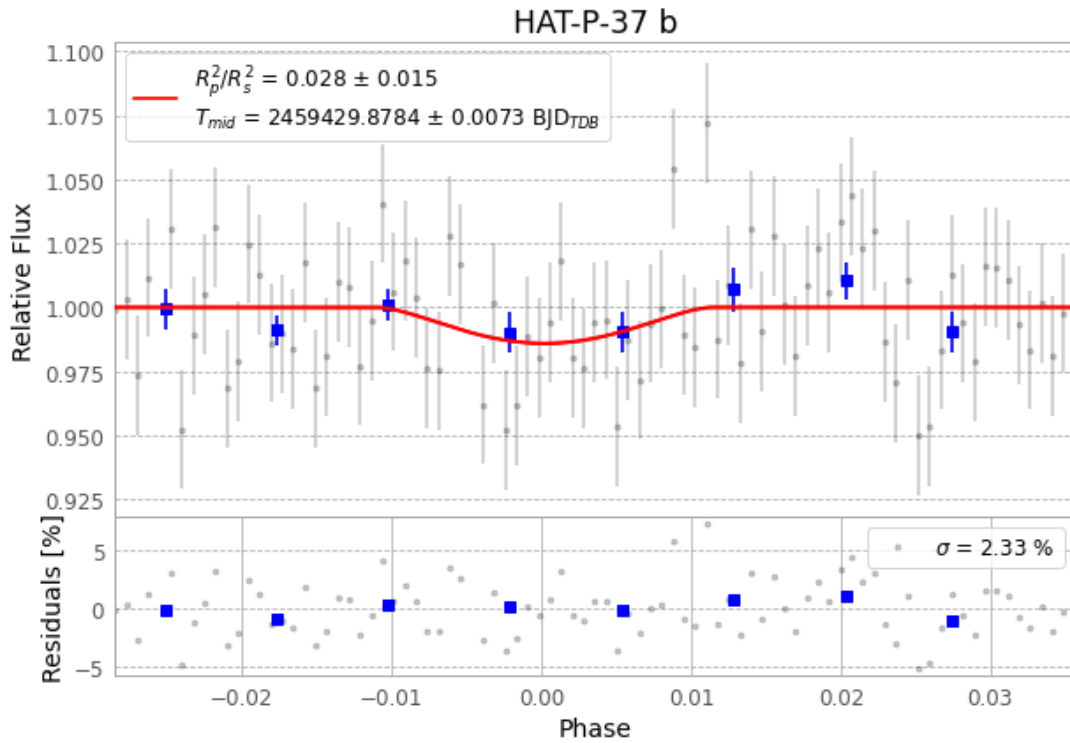


Figure 1 — FinalLightCurve_HAT-P-37 b_2021-08-03.png (SHA-256 bd9e7a36f0a6fea7...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [442, 265], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 cbad59c4243e0a28...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

87 FITS frames (57 MB), HATP-37210803070612.FITS → HATP-37210803112413.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-37-210803.log (0274441f712c...), FinalLightCurve_HAT-P-37 b_2021-08-03.png (bd9e7a36f0a6...), FinalLightCurve_HAT-P-37 b_2021-08-03.csv (18c55861e7a3...)

Compute resources

Wall time 113.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-28 00:00Z.